

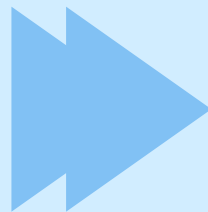
SLEEP DEPENDENT MEMORY

How Rest Shapes What We Remember

By: Sng Zhenhao



HOW DOES SLEEP BENEFIT MEMORY



“what could not be repeated at first is readily put together on the following day, and the very time which is generally thought to cause forgetfulness is found to strengthen the memory”

~Quintilian, 1st century AD

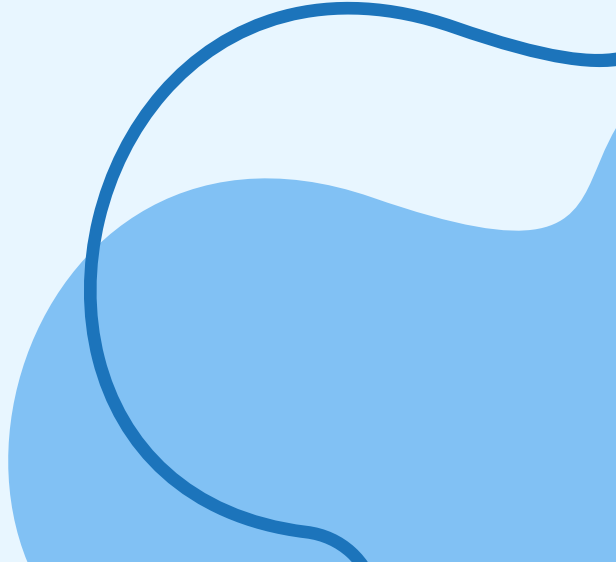


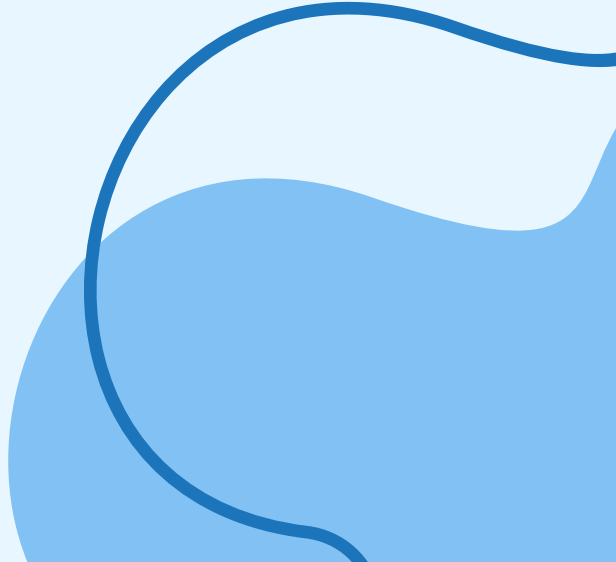
PAPERS USED:

Paper 1: Sleep-dependent memory consolidation
Robert Stickgold (2005)

Paper 2: Sleep-Dependent Learning and Memory Consolidation
Matthew P. Walker and Robert Stickgold (2004)

Paper 3: Sleep-Dependent Learning: A nap is as good as a night
Sara Mednick, Ken Nakayama & Robert Stickgold (2003)





PAPER 1: “WHAT IS HAPPENING?”

WHAT IS MEMORY?

DECLARATIVE

EPISSODIC

SEMANTIC

NON-DECLARATIVE

PRIMING

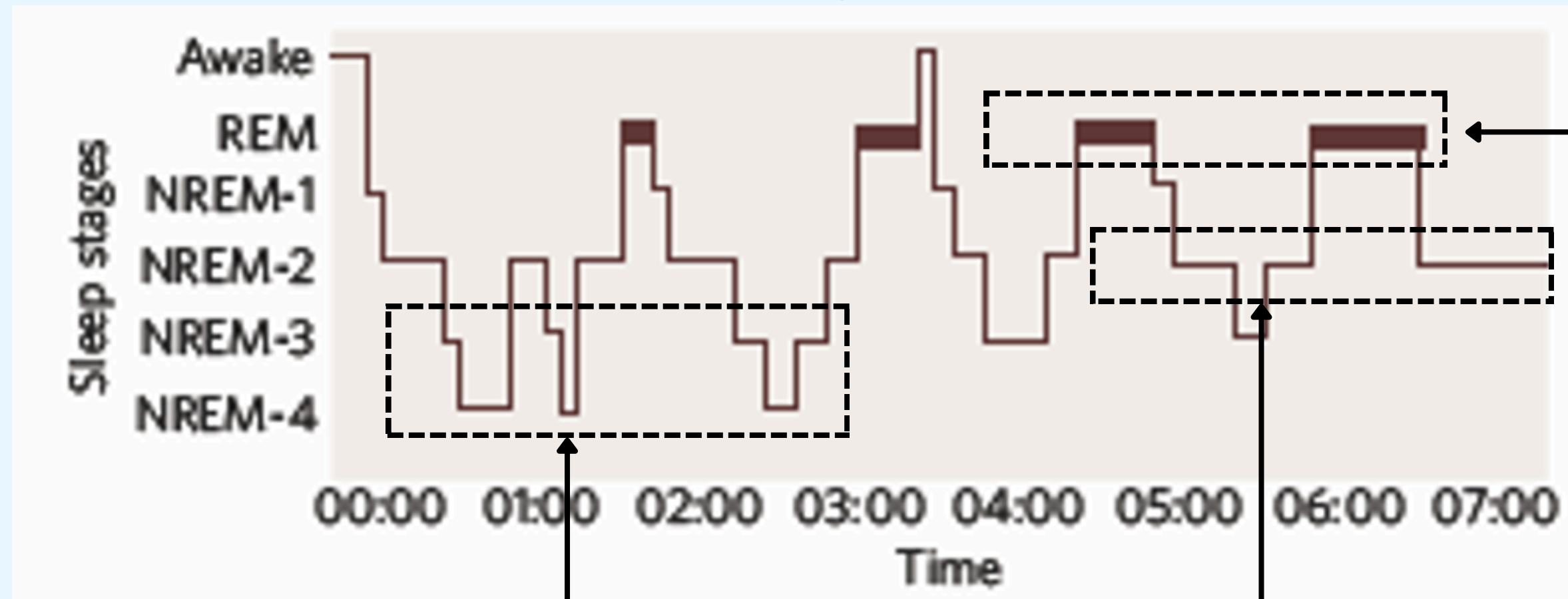
CONDITIONING

PROCEDURAL SKILLS

NON-ASSOCIATIVE

WHAT IS SLEEP

SLEEP CYCLES ROUGHLY 90 MINUTES, REPEATING 4-5 TIMES A NIGHT



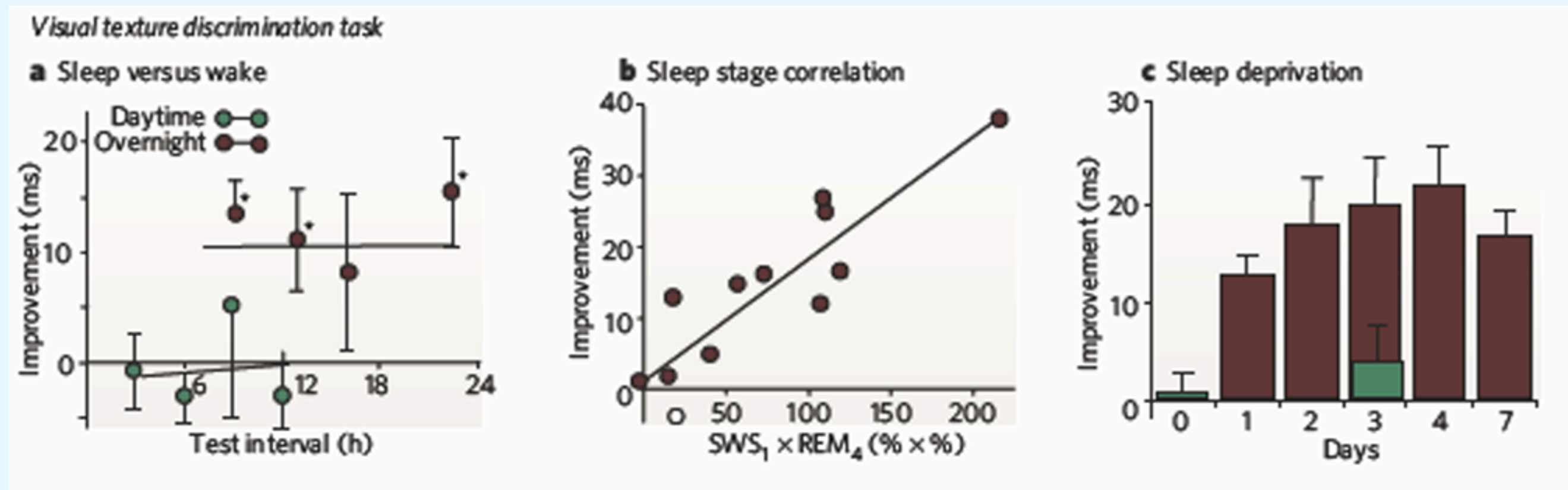
SLOW WAVE SLEEP (SWS)
[DUE TO LARGE AND SLOW OSCILLATIONS (0.5-4HZ)]

STAGE 2 NREM

REM

HOW DO THEY DETERMINE

1. Visual Discrimination Task



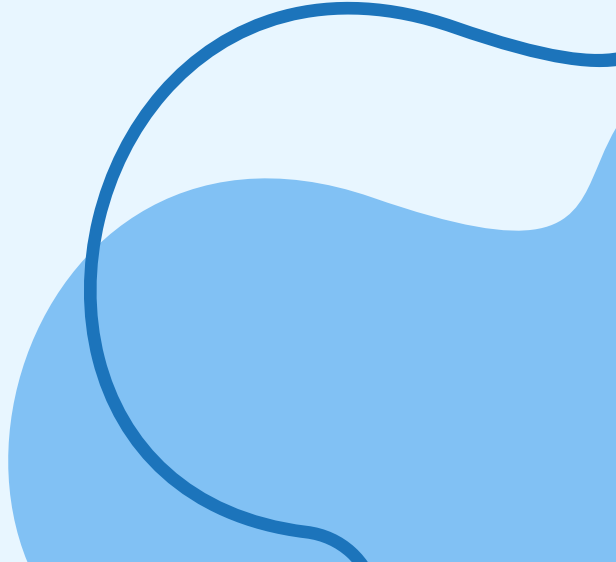


WHAT DOES THIS MEAN?

Experiment 1:

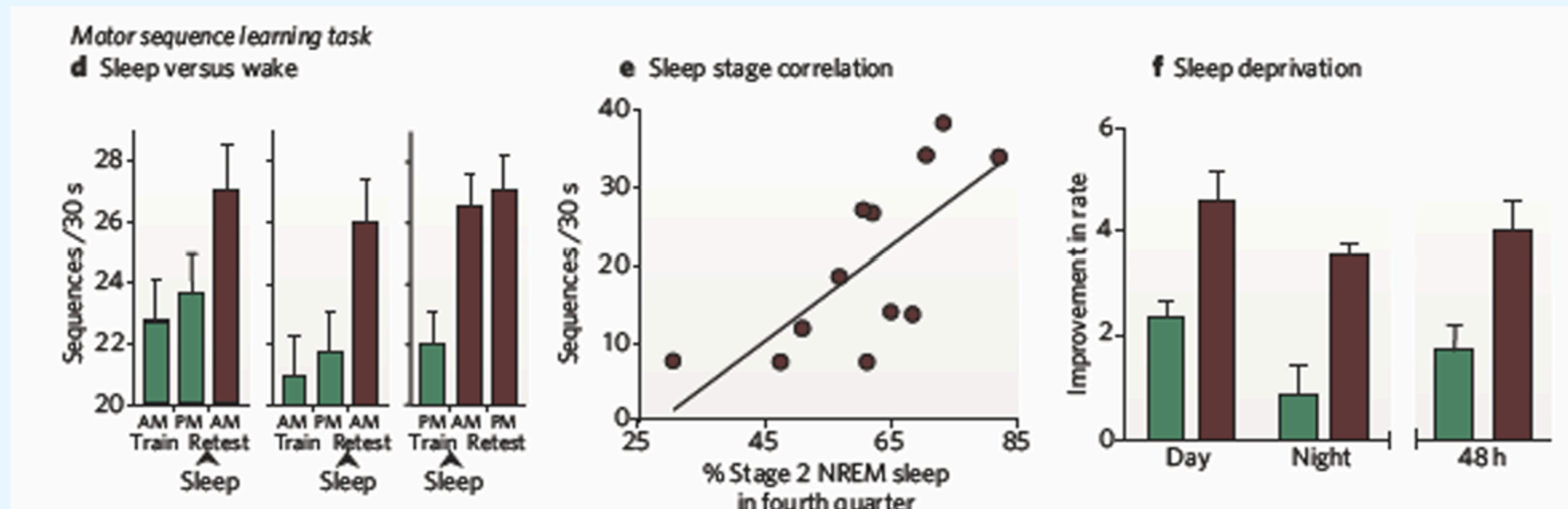
- after training, **sleep → improvement, no sleep → little to no significant improvement**
- sleep disrupted: if rem blocked, improvement. if sws blocked, no improvement

however, it was discovered that **strongest performance when tested with SWS1 x REM4.**

- importance of first night of sleep after training: if sleep deprived, later sleep does not compensate
- 

HOW DO THEY DETERMINE

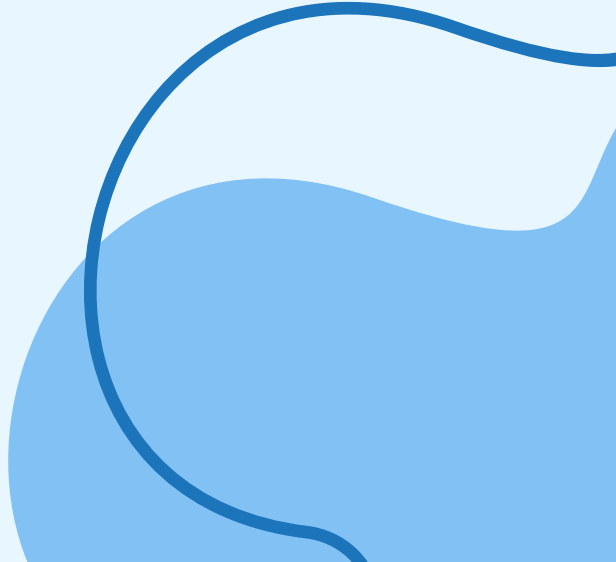
2. Motor sequence learning task

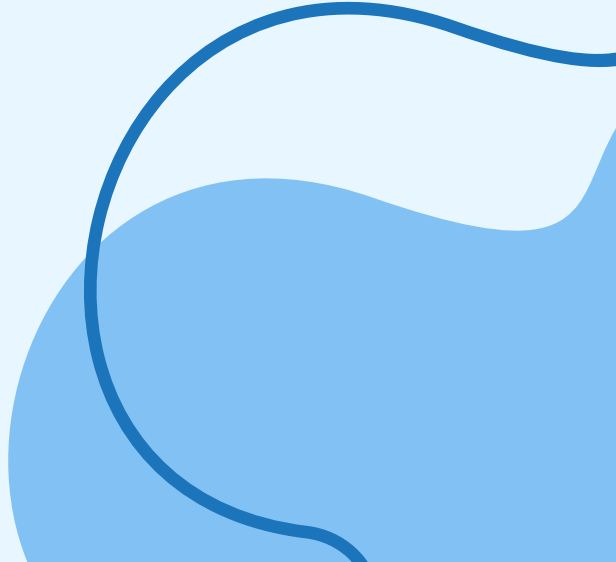




WHAT DOES THIS MEAN?

Experiment 2:

- practice improves performance but plateaus
 - staying awake does not improve much: even after 4 to 12 hours, non-significant improvement
 - However, **sleep improves performance**, about 20%
 - for finger tapping **NREM2 is more important** as compared to visual discrimination's SWS + REM → different tasks depend on different sleep stages
- 



PAPER 2: “HOW IS IT HAPPENING?”

DIFFERENT BRAIN WAVE PATTERNS ARE EXHIBITED

- Stage 2 NREM contains sleep spindles and K-complexes (EEG); motor sequence improvement has been linked to Stage 2 NREM
- Stage 1 refers to light sleep, does not contain spindles
- Deep NREM: slow, synchronized brain waves
- REM: faster, desynchronized brain activity
- Wake: faster, desynchronized brain activity

Chemistry	REM	WAKE	Non REM
acetylcholine	high	high	low
serotonin	very low	high	low
noradrenaline	very low	high	low

DIFFERENT BRAIN WAVE PATTERNS ARE EXHIBITED

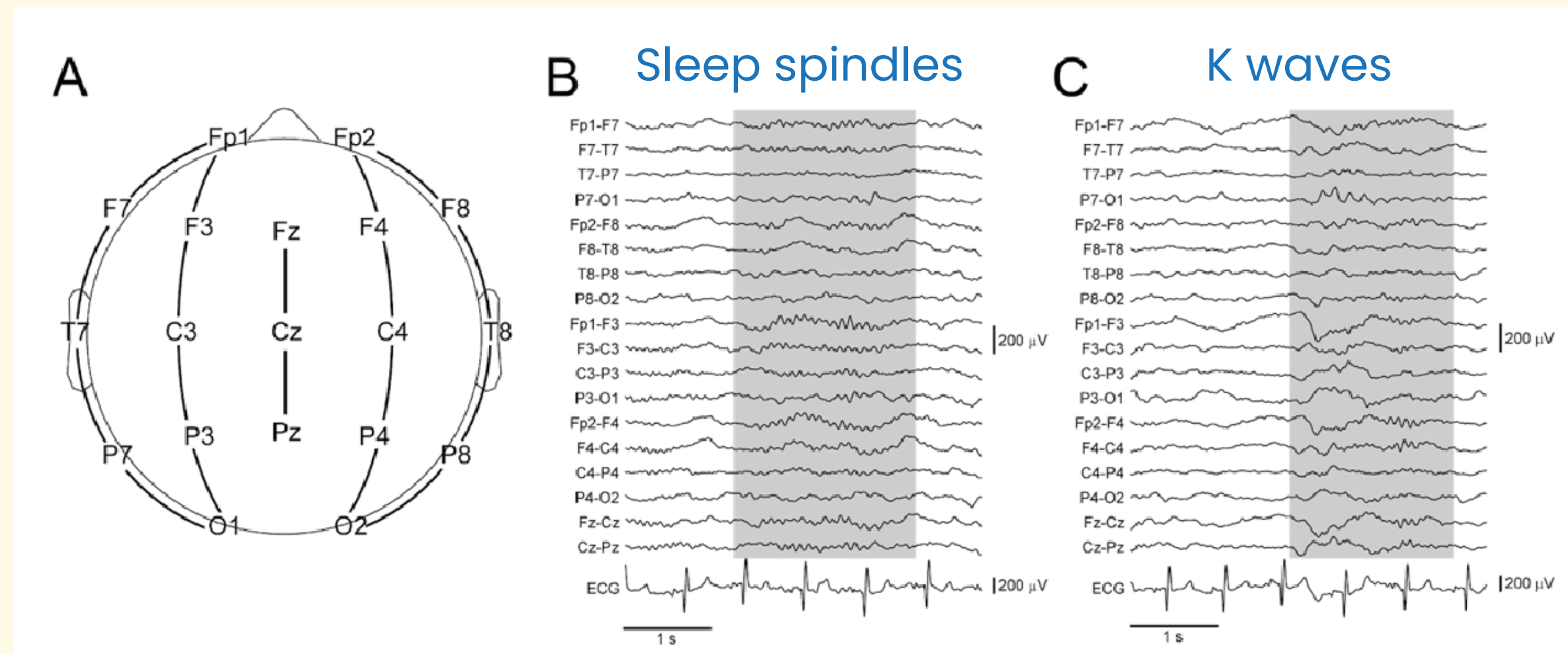
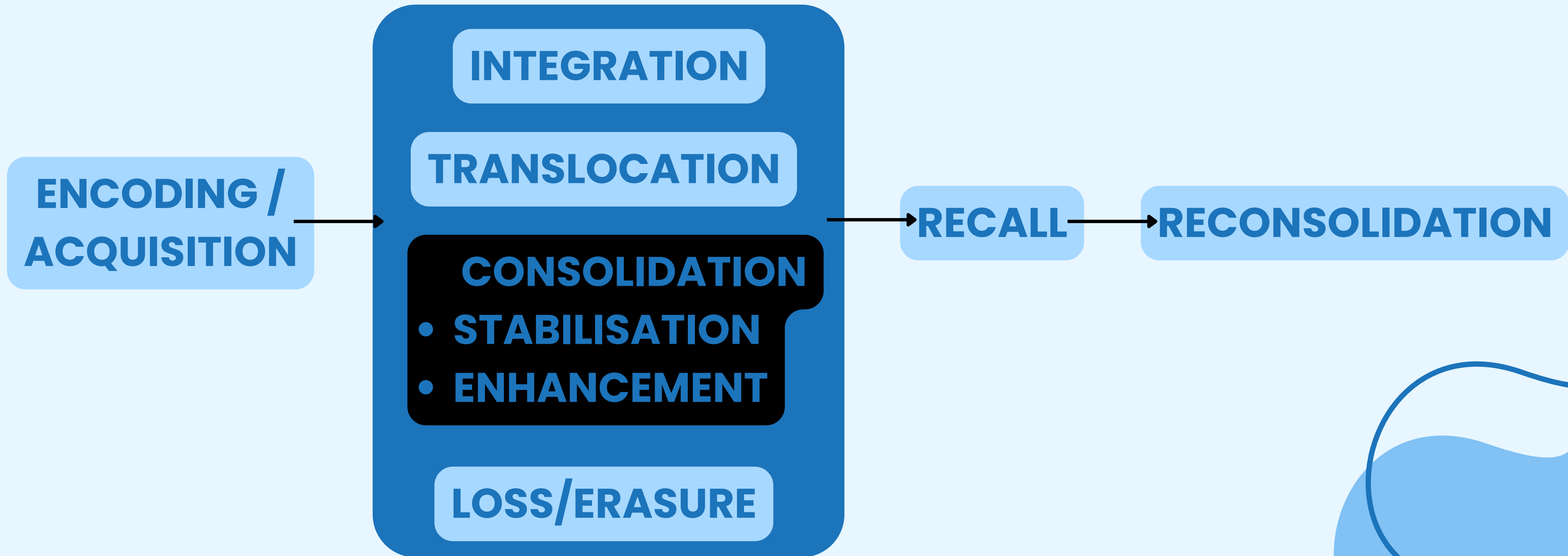


Figure 1: EEG system and visual detection of sleep waveforms. A) EEG 10-20 system in double-banana montage. B) Representative example of an EEG spindle. C) Representative example of a spontaneous K-complex, including the initial vertex and its trailing spindle.

MEMORY STAGES



WHY MEMORY TYPE MATTERS

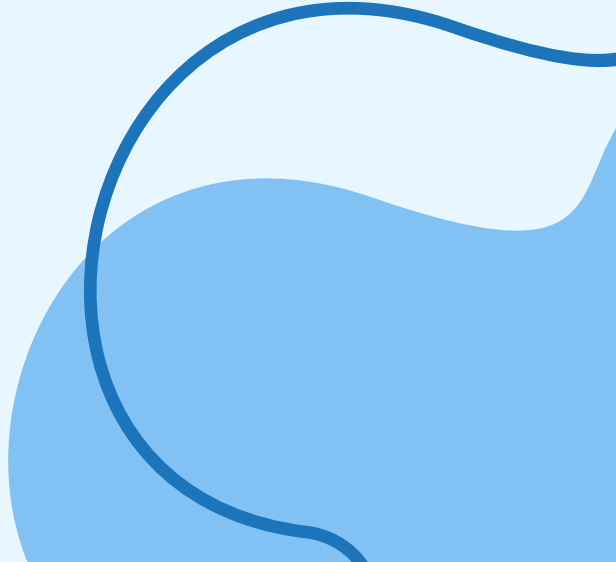
Declarative memory	Non-declarative memory
Facts and events	Skills, habits, perceptual learning
“Knowing what”	“Knowing how”
Hippocampus / medial temporal lobe important (hippocampus may provide a retrieval code for neocortically stored information)	Less dependent on medial temporal lobe
Evidence with sleep is more mixed, especially for simple neutral tasks	Stronger evidence in the papers: visual discrimination, motor sequence

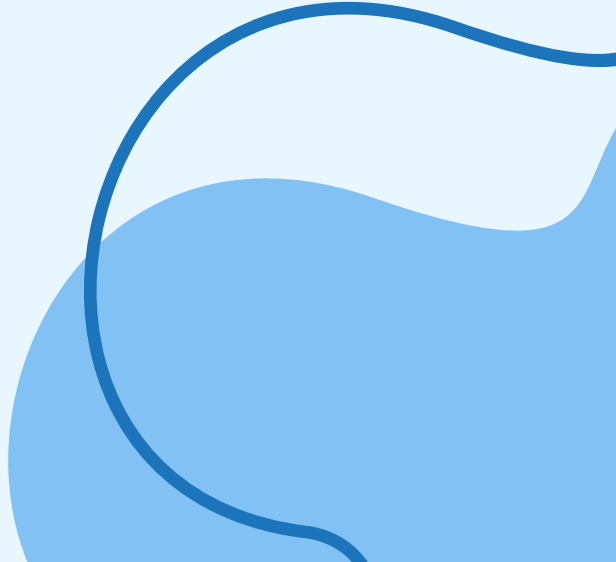


WHAT DOES THIS MEAN?

Sleep is not one uniform thing. SWS, Stage 2 NREM, and REM have different brain rhythms and neurochemistry, so they may support different memory processes

Different memory systems go through **different consolidation stages**, and **different sleep stages** may support different parts of that process.



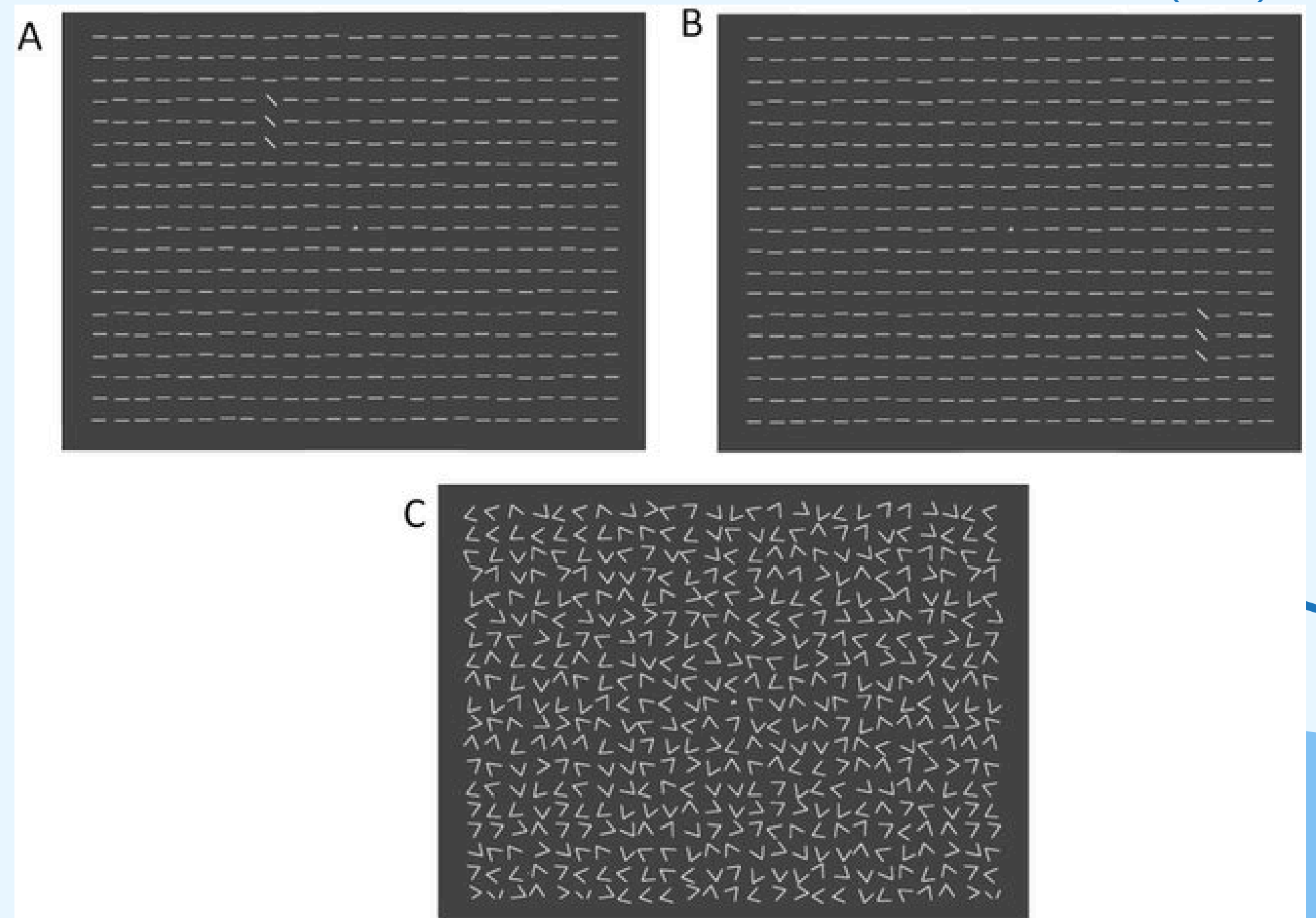


PAPER 3: “SO WHAT?”

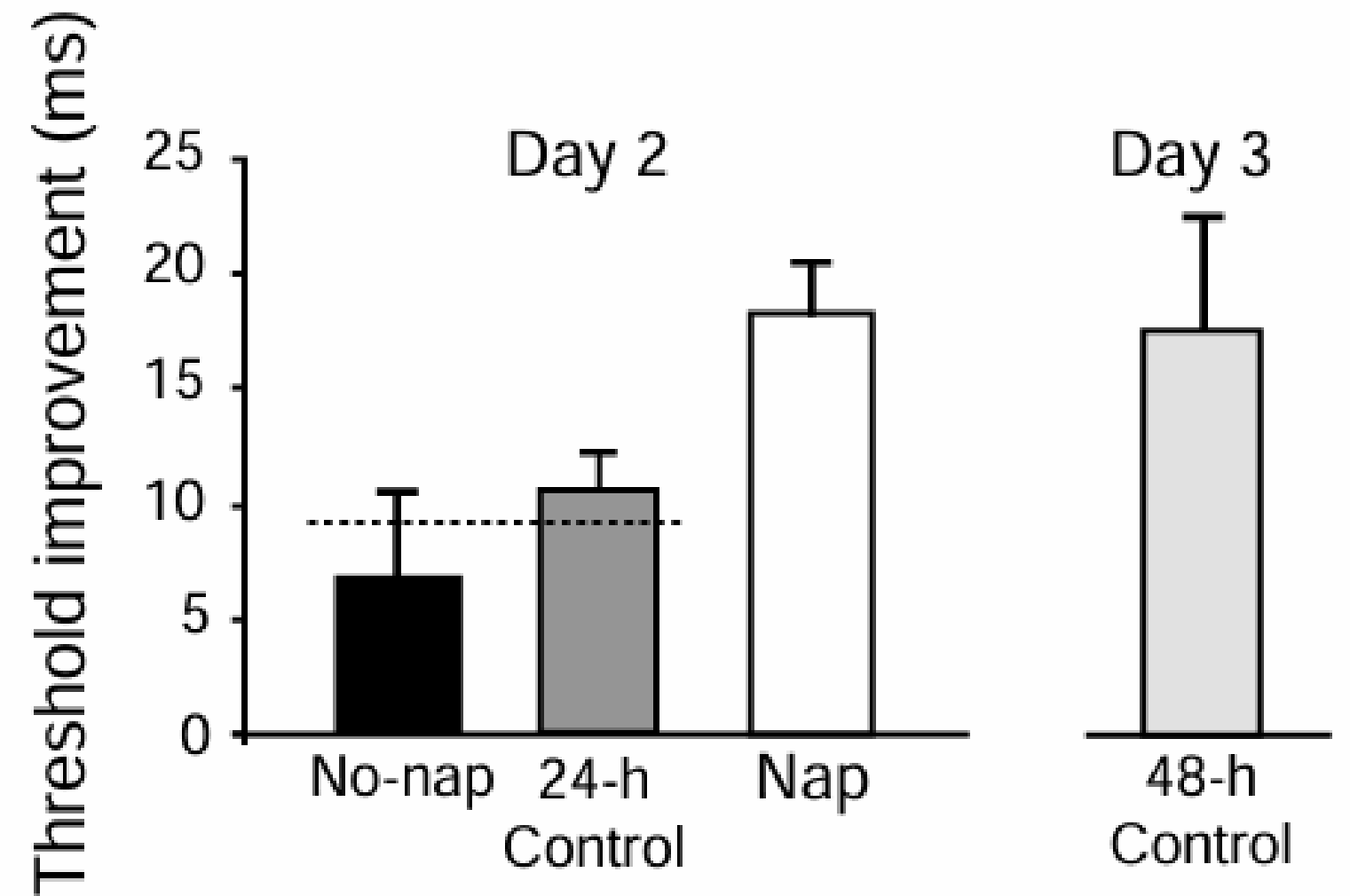
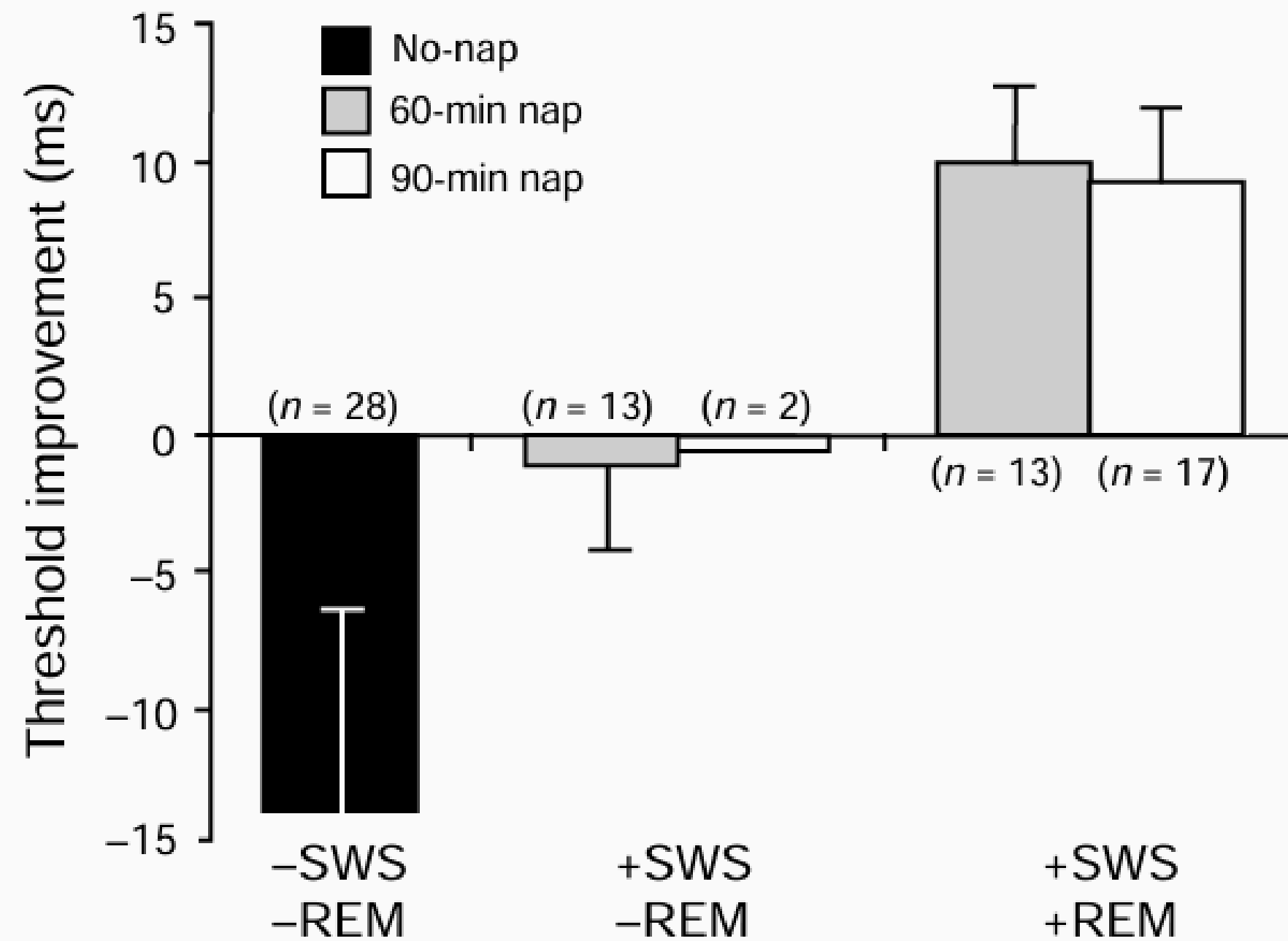
THE EXPERIMENT

- 2 groups: nap (60 – 90 mins) vs no nap
- Textual discrimination task (tdt) training at 9am, first retest at 7pm, subsequent testing at 9am the next day
- Task sequence: fixation → target(A/B) → blank/ISI → mask (C) → response

TEXTURE DISCRIMINATION TASK SCHEMATIC.
ADAPTED FROM HARRIS AND SAGI (2018)



RESULTS





RESULTS

Nap containing both SWS and REM produces the same magnitude of learning improvement as a full night of sleep on the texture discrimination task

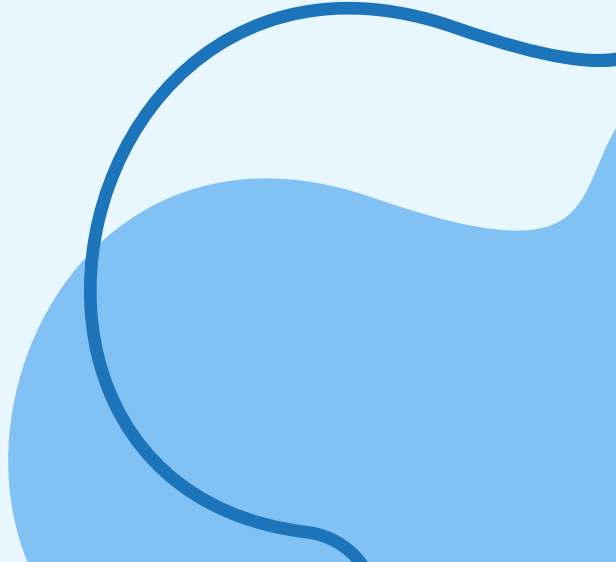
SWS alone in a nap = reverses deterioration

not timing dependent, 60 min nappers grouped with 90 min if there is REM

Improvements were additive, nap group then slept the night and improved again, ending up with double the benefit of the no-nap group over 24 hours

The nap + one night of sleep produced about as much improvement as two nights of sleep without a nap.

(*This study applies to non-declarative memories)



CONCLUSION



**SLEEP IS NOT THE ENEMY OF LEARNING. IT MAY BE
PART OF HOW LEARNING COMPLETES ITSELF.**

SLEEP WELL, REMEMBER MORE

Prioritize rest for a sharper, stronger mind.





ANNEX

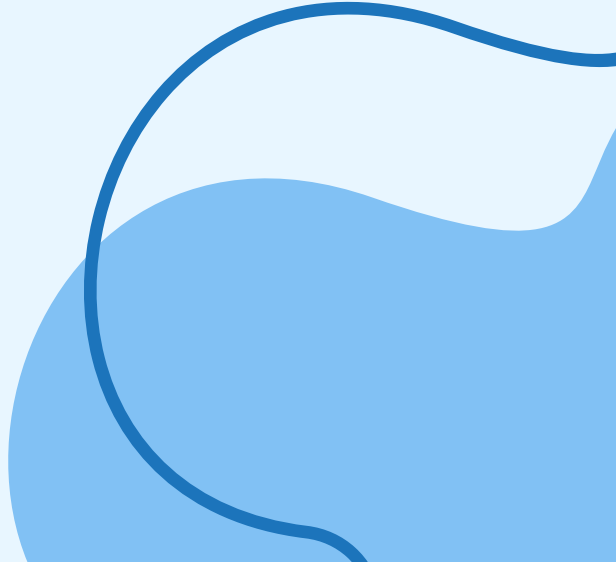
CAVEAT ON SLEEP DEPENDENT

SLEEP-LIKE THINGS CAN HAPPEN DURING WAKE:

- HALLUCINATIONS
- DAYDREAM-LIKE IMAGERY

WAKE-LIKE THINGS CAN HAPPEN DURING SLEEP:

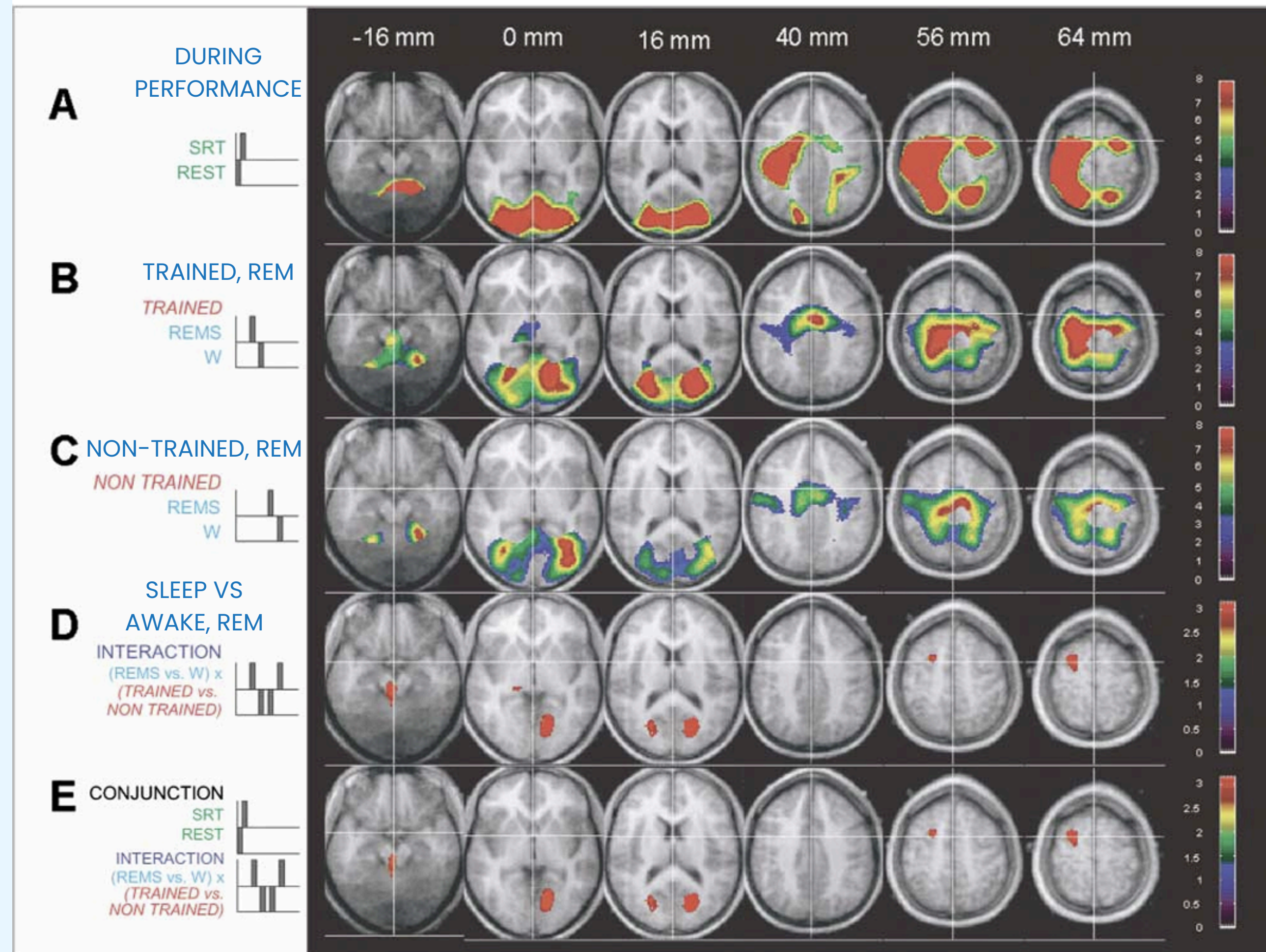
- SLEEPWALKING
- SOME BRAIN ACTIVITY DURING REM LOOKS AWAKE-LIKE

- SLEEP AND WAKE ARE NOT DIFFERENT BOXES
 - “SLEEP-DEPENDENT” = SLEEP IS THE MAIN CONDITION WHERE THE EFFECT APPEARS, NOT NECESSARILY THE ONLY POSSIBLE BRAIN STATE WHERE RELATED MECHANISMS EXIST.
- 

PET SCAN

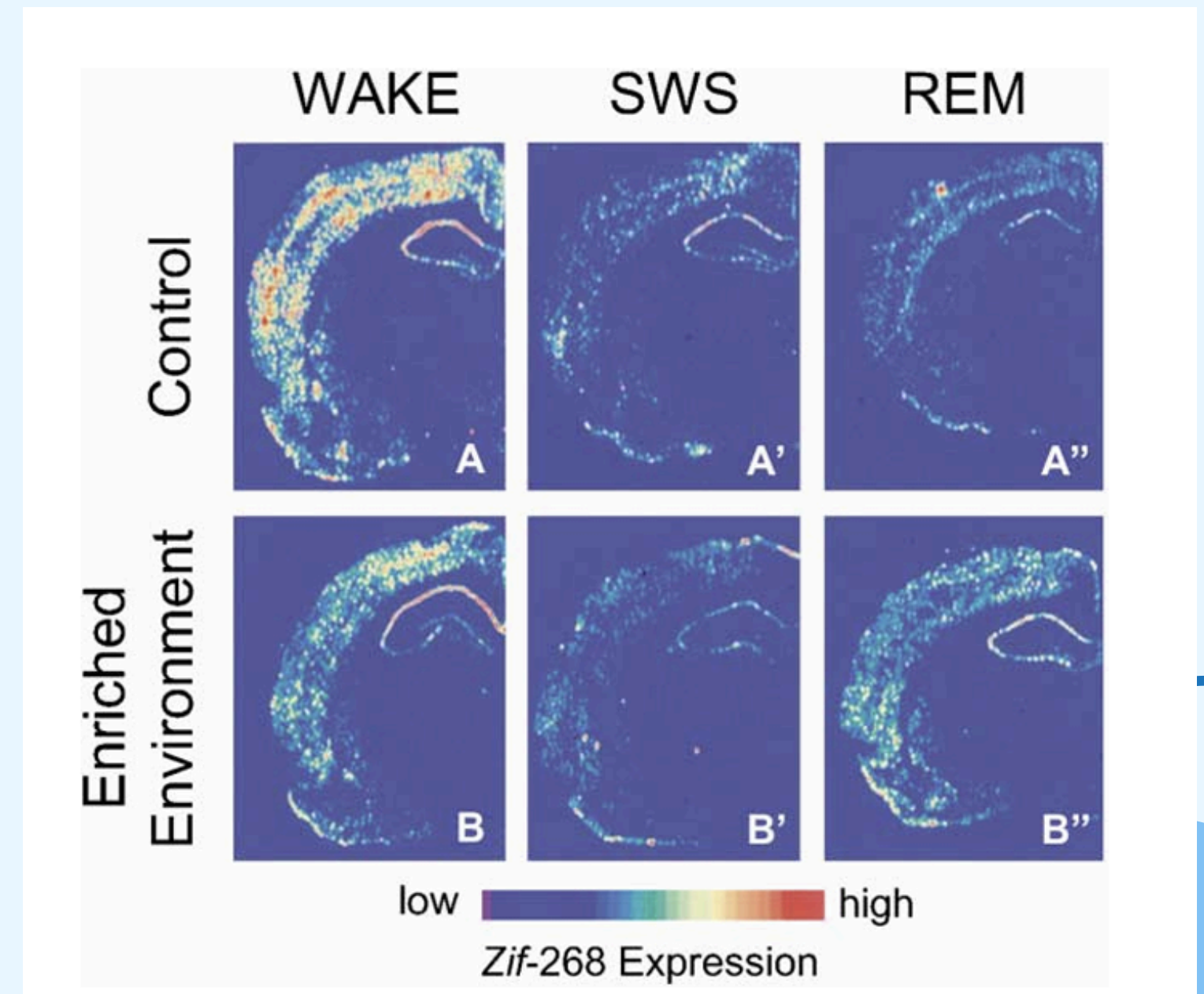
D: What REM activity is specifically stronger in trained people (difference btw training vs non training)

E: Which brain areas overlap between task performance and REM reactivation (A and B's intersection)



MEANINGFUL MEMORY

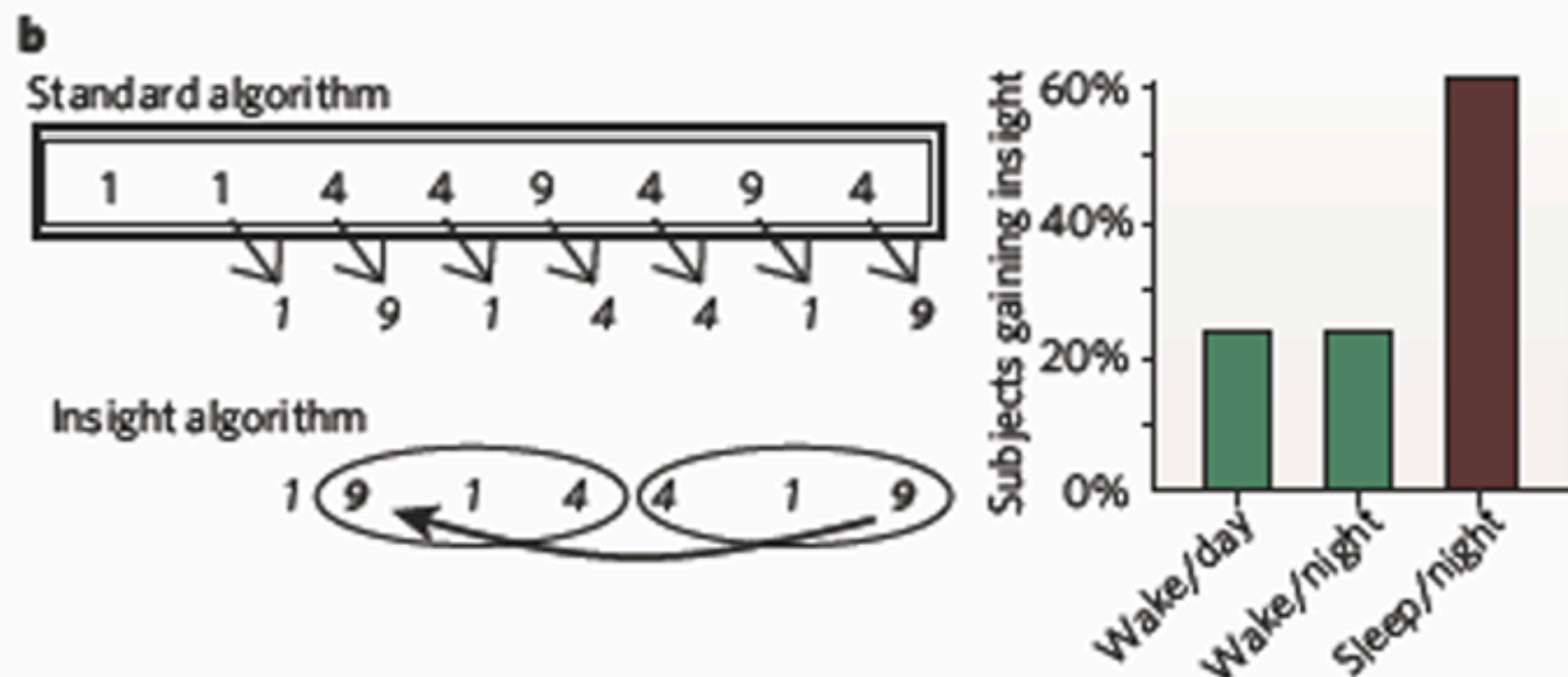
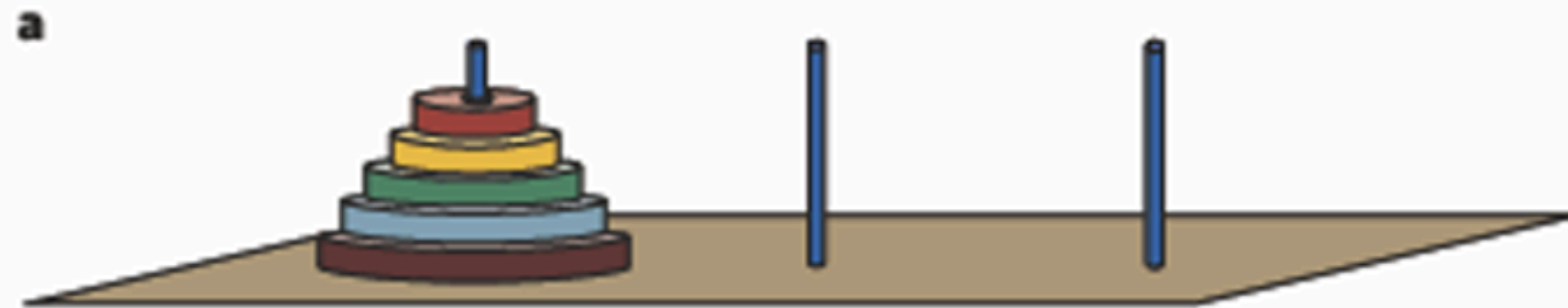
- zif-268 gene to neural plasticity.
- rats exposed to rich sensorimotor environment (stimulating their senses & explore)
- during rem sleep, gene expression increased
- meaningful waking experience → later during REM sleep, a plasticity-related gene increased → therefore REM may be processing that prior experience
- gene activation is based on stimulus prior to sleeping, so REM sleep supports experience dependent plasticity
- tldr; REM sleep may support memory after meaningful experiences



PGO WAVES

- In animal models, it was found that sleep benefits hippocampus dependent memory. **PGO waves** = bursts of brain activity during REM sleep, involving the **pons**, **lateral geniculate nucleus**, and **occipital cortex**. But p waves in rats. So they did an experiment on avoidance training on rats, and found that after doing that they required more REM sleep, then p waves density increased, and those with more p- waves activity led to better memory retention. -> p / pgo waves instead of rem sleep instead
- By artificially inducing PGO waves, cholinergically driven brain activity rather than REM per se is necessary for the sleep-dependent enhancement of this consolidation.

TOWER OF HANOI



Those who slept between training and testing were roughly three times more likely to discover a hidden shortcut algorithm, suggesting sleep may also support insight and restructuring of memories.

DECLARATIVE MEMORIES

Some found that **sleep benefits declarative memory while others don't.**

For simple, neutral declarative memories, like memorizing random word pairs with no emotion or meaning, the evidence is mixed. (dog-leaf [entirely new connection → mixed effect] vs dog-bone [pre-existing/meaningful → improved memory]).

But for complex declarative memories, both SWS and REM sleep help consolidation.

Task difficulty and emotional importance affected by sleep.

- SWS may support stabilization of hippocampal memories
- REM may support emotional/associative/flexible processing

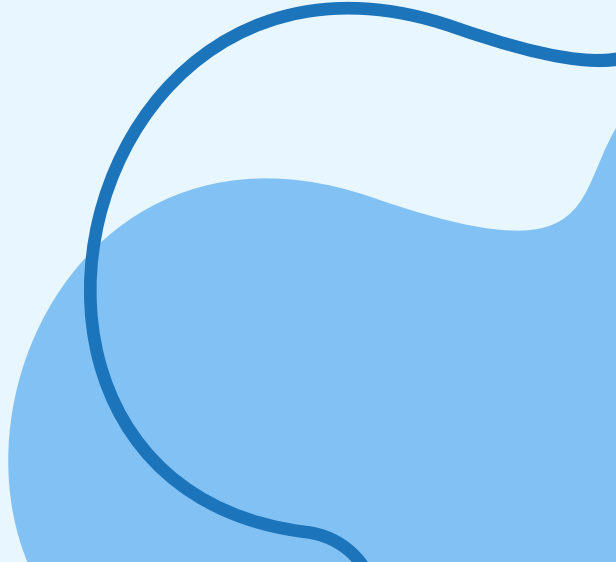
Key limitation: many studies only test recall/recognition and not other post encoding stages

The neocortex stores lots of the actual information, but the hippocampus helps organise/retrieve it in the right time/order/context.



SLEEP DEPENDENT LEARNING

Thing to note: Sleep dependent learning is a separate training process from initial stabilization and practice dependent acquisition. Also, first night of sleep after acquiring new skill is important but sleeping during the day helps too. Fischer found a correlation with **REM sleep**, while Walker found stage 2 NREM. The authors say this discrepancy is unresolved.





CITATIONS

- MEDNICK, S. C., NAKAYAMA, K., & STICKGOLD, R. (2003). SLEEP-DEPENDENT LEARNING: A NAP IS AS GOOD AS A NIGHT. NATURE NEUROSCIENCE, 6(7), 697–698. [HTTPS://DOI.ORG/10.1038/NN1078](https://doi.org/10.1038/NN1078)
 - STICKGOLD, R. (2005). SLEEP-DEPENDENT MEMORY CONSOLIDATION. NATURE, 437, 1272–1278. [HTTPS://DOI.ORG/10.1038/NATURE04286](https://doi.org/10.1038/NATURE04286)
 - WALKER, M. P., & STICKGOLD, R. (2004). SLEEP-DEPENDENT LEARNING AND MEMORY CONSOLIDATION. NEURON, 44(1), 121–133.
 - HARRIS, H., & SAGI, D. (2018). VISUAL LEARNING WITH REDUCED ADAPTATION IS ECCENTRICITY-SPECIFIC. SCIENTIFIC REPORTS, 8, 608. [HTTPS://DOI.ORG/10.1038/S41598-017-18824-7](https://doi.org/10.1038/s41598-017-18824-7)
- 